

Midterm assignment, Autumn 2025

02429 Analysis of correlated data: Mixed linear models

October 9, 2025

To be uploaded to dtulearn on the 23rd of October, latest at 8:00 am, and to hand in a printed version in class.

To be solved in groups of 2 to 3 students, groups of size 1 are not accepted.

The report is limited to a total of 25 pages plus those for the R-code appendix. A maximum of 12 pages is for the main text. The R-code should be added in an appendix and not as an independent file.

Relevant figures and tables should be included in the main text, the rest in an appendix.

Substance is more important than style, but please remember: use no more than four significant digits and fewer if possible. Use a good font size for numbers and labels in the figures. Prioritize the length of the different sections (exploratory analysis, modelling,..., conclusions) in the text.

Preamble

Perform the statistical analyses and present your analyses and conclusions.

- The R code should be neat, readable, and commented.
- All figures and tables included in the report must be referred to and discussed in the text and should be numbered and have an informative legend.
- There should be a minimum of R code/output in the main body of the report. In appendices, R output is allowed.
- Remember that much of Statistics is about quantifying and describing variability and uncertainty: When presenting parameter estimates, remember to include confidence intervals.
- Present the mathematical expression of the initial and final model.
- Think about the reader when writing your report. The readers are those in the class. You should include enough detail and explanation in the body of your report and allow extra material in an appendix that supports your numerical results:
 - a) to allow a statistician with no knowledge of R to replicate your analysis and validate your results using other statistical software *and*
 - b) for someone with little or no knowledge of Statistics to understand the overall purpose and conclusions of your report in the context of the data.

- Include an exploratory analysis, modelling, results, answer to the research question, conclusion(s) and relevant comments.
- Do not forget to include: `ranova`, `anova/drop1`, `summary`, `residplots`, for the final model, either in the main text or in an appendix. These will allow us to verify results.

Exercise I. Bang & Olufsen tests

Data

In a sensory experiment, televisions from Bang & Olufsen were evaluated by 8 assessors with respect to the colour saturation. In the experiment, the combinations of 3 different pictures and 3 different television settings were investigated. The experiment was repeated once, meaning that we have 2 replicates. The purpose of the experiment was to compare the pictures, the television settings, and their combination in terms of colour saturation.

The variable **Assessor** indicates the assessor (1-8), **TVset** the television setting (TV1,TV2,TV3), **Picture** the picture (1-3), **Repeat** the replicate (0,1), and **Coloursaturation** denotes the colour saturation score given by the assessor. The data is available in the file `TV2.csv`.

Perform a statistical analysis in which you investigate the effect of picture and television setting on the colour saturation. The statistical inference should not only be valid for the 8 assessors in the experiment (which we assume were chosen at random from a larger population of possible assessors).

Exercise II

In some applications of subsampling, the major classes have fixed effects to be estimated (sires). An instance is an evaluation of the breeding value of a set of five sires in raising pigs. Each sire is mated to a random group of dams, each mating producing a litter of pigs whose characteristics are the criterion. The table below shows the data of an example with 2 dams for each sire and two pigs chosen from each litter.

In this setting, the sires factor is considered fixed, whereas Dam and Pig Gains random. Fit a model to investigate whether the sires have breeding values that are statistically different, and if so, give the estimated values.

Report the expression or formula of the model you used to justify your answer, the fitting and relevant numerical and graphical results of the model, and comment on your results.

TABLE 13.9.1
AVERAGE DAILY GAIN OF TWO PIGS OF EACH LITTER

Sire	Dam	Pig Gains		Sums	
1	1	2.77	2.38	5.15	10.67
	2	2.58	2.94	5.52	
2	1	2.28	2.22	4.50	10.12
	2	3.01	2.61	5.62	
3	1	2.36	2.71	5.07	10.53
	2	2.72	2.74	5.46	
4	1	2.87	2.46	5.33	9.88
	2	2.31	2.24	4.55	
5	1	2.74	2.56	5.30	10.28
	2	2.50	2.48	4.98	
					51.48

Exercise III. Junior School Project

The dataset `df_wide.csv`, which is a subset of `jsp.csv`, consists of 1093 pupils in 49 primary (elementary) schools in inner London, and it is part of the 'Junior School Project' (JSP). We consider two measurement occasions: the first when the pupils were in their fourth year of schooling, and three years later in their final year of primary school, seventh year. A smaller dataset has been described and analyzed in "Multilevel Statistical Models", H. Harvey, second edition.

We use the scores from mathematics tests administered on these two occasions together with information collected on the social background of the pupils, a general knowledge test called raven's, and gender.

Fit a hierarchical/multilevel model where:

- Response variable: `maths.1`
- Explanatory variables: `maths.0`, `ravens`, `sex`, `sc`
- Random factors: `school`, `class`

NB: Variable `class` in the data set takes values "1", "2", "3", "4". the last two categories are present in only few schools. You may consider to random effects: only schools, only class, and class nested in school. And select one option y your final model. The factor social class takes too many categories, and unfortunately we do not have the labels for the categories.